Arista Access Node MAR-ACC-49 Marathahalli Core Routing Protocol Flap Resolution SOP (ALARM_9815)

Node: MAR-ACC-49 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_9815

Description

On the Arista Access Node MAR-ACC-49, situated at the Marathahalli Core site, a routing protocol flap occurs due to a BGP route churn, OSPF adjacency loss, or ISIS LSP flooding. This phenomenon is primarily attributed to the dynamic nature of routing protocols, which continuously seek optimal paths for packet forwarding. As the network topology evolves, BGP routes may fluctuate, causing a churn in the routing table, while OSPF adjacency loss might occur due to network instability or misconfiguration. Similarly, ISIS LSP flooding can result from excessive network changes or hardware-related issues. This SOP aims to troubleshoot and resolve the underlying causes of the routing protocol flap on the specified Arista Access Node, leveraging the vendor-specific CLI dialect to navigate the control and data planes.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Marathahalli Core site to physically inspect the equipment and network connections.

- Verify the status of redundant links to determine if there are any issues with backup connectivity.

- Check the system logs for any recent firmware upgrades, configuration changes, or other relevant events that might have contributed to the routing protocol flap.

Potential Root Causes

Cause	Probability
BGP route churn due to network topology changes	High
OSPF adjacency loss caused by network instability	Medium
ISIS LSP flooding resulting from excessive network changes	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is XX, main routing table version is XX, Description: This command displays a summary of the BGP routing table, providing information on the number of routes, neighbors, and updates. The output will help identify any BGP route churn or instability.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command shows a list of OSPF neighbors, their state, and the interface they are connected
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```
through. The output will help identify any OSPF adjacency loss or
network instability.

Command: show isis database, Expected Output: ISIS LSP Database,
Description: This command displays the ISIS link-state database,
providing information on LSPs, levels, and metrics. The output will
help identify any ISIS LSP flooding or excessive network changes.
```

Resolution Steps

Step 1: Step: Verify BGP routing table stability, Command: show ip bgp summary, Description: Check the BGP routing table for any flapping routes or instability. If issues are found, refer to the BGP configuration and ensure that the network topology changes are properly reflected in the routing table.

Step 2: Step: Re-establish OSPF adjacency, Command: clear ip ospf process, Description: Clear the OSPF process and re-establish adjacency with neighboring routers. This will help resolve OSPF adjacency loss and network instability issues.

Step 3: Step: Suppress ISIS LSP flooding, Command: isis database flush, Description: Flush the ISIS LSP database and suppress any flooding caused by excessive network changes. This will help resolve ISIS LSP flooding and network instability issues.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is
XX, main routing table version is XX, Description: Verify that the BGP
routing table is stable and there are no flapping routes or
instability.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: Verify that OSPF adjacency
has been re-established and there are no network instability issues.

Command: show isis database, Expected Output: ISIS LSP Database,
Description: Verify that the ISIS LSP database is stable and there is
no flooding caused by excessive network changes.
```

Roll-back Procedures

- Step: Revert BGP configuration changes, Command: show ip bgp conf, Description: Revert any BGP configuration changes made during the resolution process to ensure that the network returns to its original state.

- Step: Re-establish OSPF adjacency, Command: clear ip ospf process, Description: Re-establish OSPF adjacency with neighboring routers to ensure that the network returns to its original state.

- Step: Restore ISIS LSP database, Command: isis database restore, Description: Restore the ISIS LSP database to its original state to ensure that the network returns to its original state.